

Fundamentals of Cloud Computing and IoT (SEng7232)

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Chapter - 1
Introduction

Cloud VS Computing

cloud

- a visible collection of particles of water or ice suspended in the air.
- Relating to or doing business on the Internet.



computing

- The process of utilizing computer technology to complete a task.
- The act of calculating or reckoning.
- Computing is the study of how computers and computer systems works .





Introduction

- With traditional desktop computing, we can run copies of software programs on our own computer.
- **The documents we create are stored on our own pc.**
- Although documents can be accessed from other computers on the network, they can't be accessed by computers outside the network, i.e. PC-centric.
- **With cloud computing, the software programs one use aren't run from one's personal computer, but are rather stored on servers accessed via the Internet.**

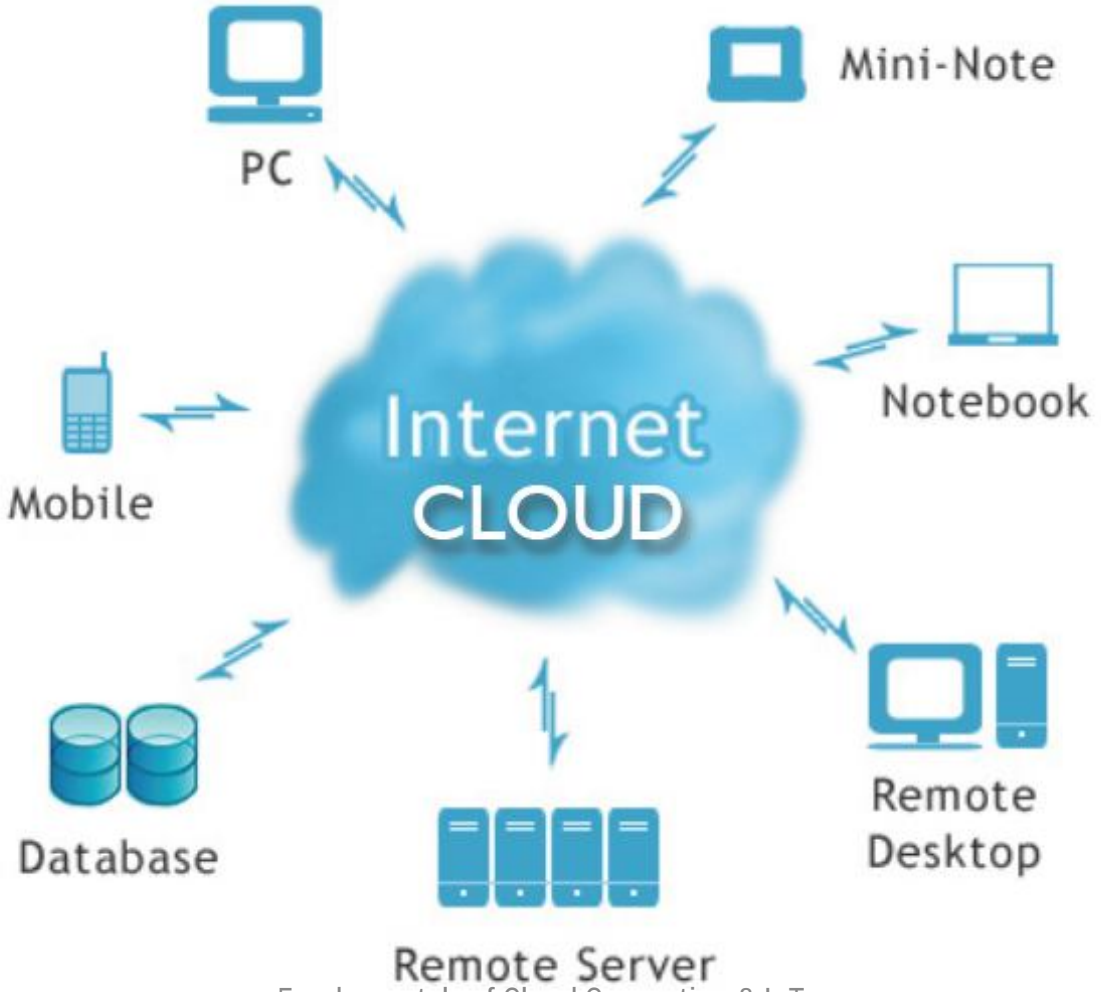
Cont. ...

- If a computer crashes, the software is still available for others to use.
- Anyone with permission can not only access the documents, but can also edit and collaborate on those documents in real time.
- **Unlike traditional computing, this cloud computing model isn't PC-centric, it's document-centric.**

What is Cloud Computing?

- It is a platform on internet.
- A type of computing that relies on sharing physical resources rather than having local server.
- A type of internet based computing where server, storage and application are deliver through internet.
- Cloud computing is a kind of **grid computing.**

Everything on the Clouds



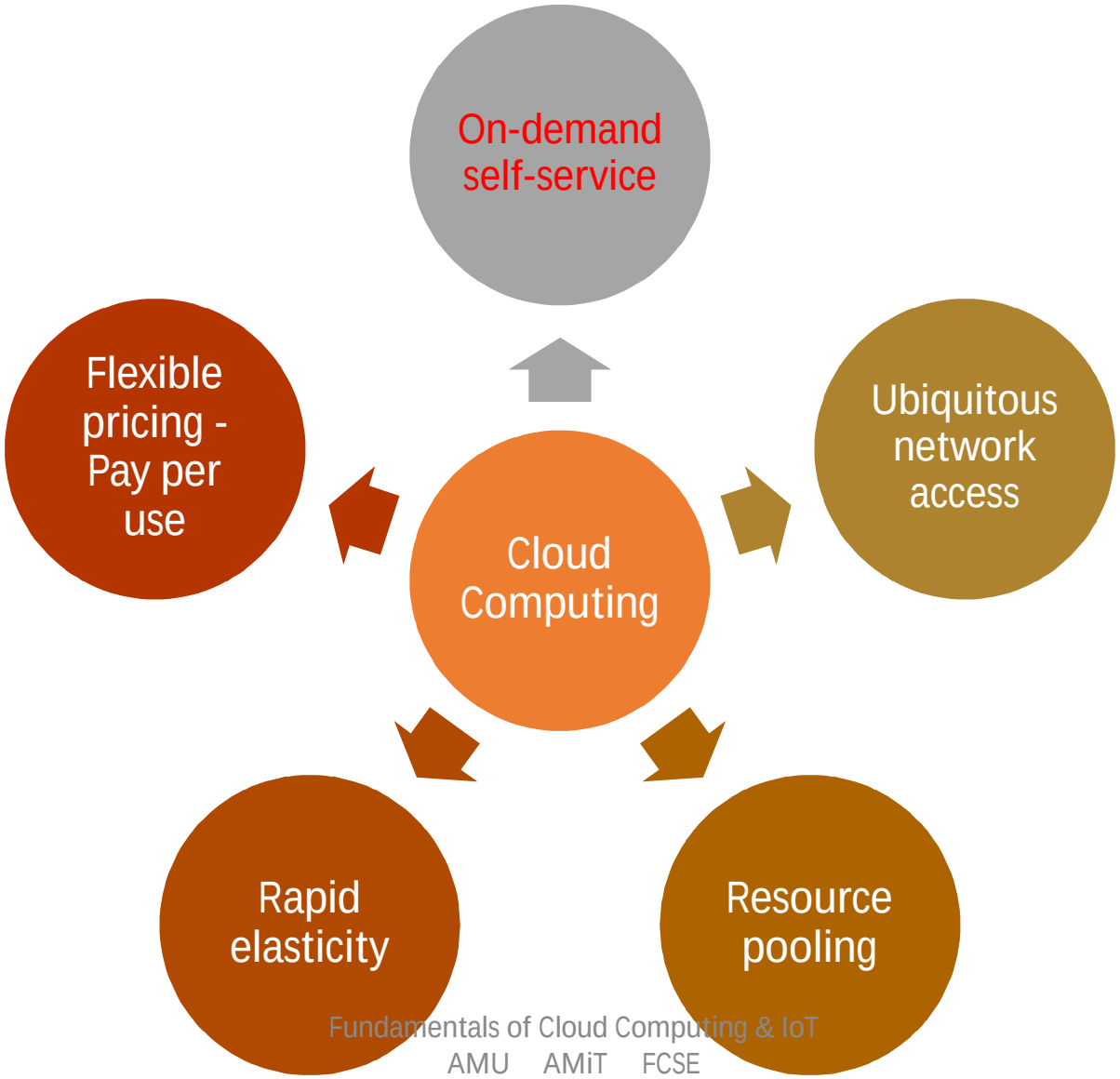
Why cloud computing

- Cloud Computing Is User Centric
- Cloud Computing Is Task-Centric
- Cloud Computing Is Powerful
- Cloud Computing Is Accessible
- Cloud Computing Is Intelligent
- Cloud Computing Is Programmable

History and Origins

- The term “cloud” is used as a metaphor (symbol) for the internet, based on the cloud drawing used in the past to represent the telephone network, and later to depict the internet in computer network diagrams as an abstraction of the underlying infrastructure it represents .
- **The concept of cloud computing introduced in 1960s, when jhon mccarthy coined that “computation may someday be organized as a public utility.”**

Characteristics of Cloud computing



Cloud Architecture

- Individual users connect to the cloud from their own personal computers or portable devices, over the Internet. To these individual users, the cloud is seen as a single application, device, or document.
- The hardware in the cloud (and the operating system that manages the hardware connections) is invisible.

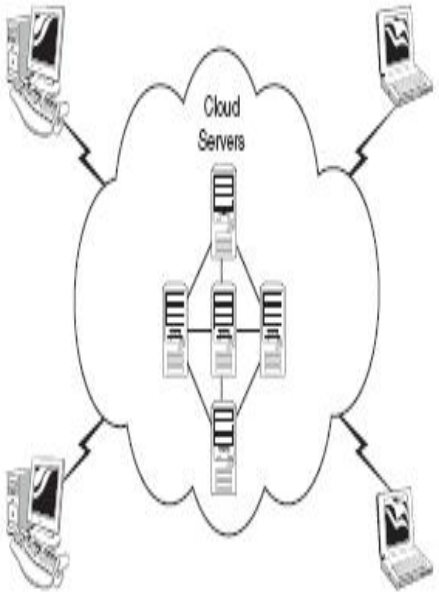


FIGURE 1.1
How users connect to the cloud.

Cloud Service Models

cloud computing service providers offer their services according three fundamental models:

1. Infrastructure as a service (Iaas),
2. Platform as a service (Paas),and
3. Software as a service(Saas)

Infrastructure as a service (Iaas),

- In this most basic cloud service model,cloud providers offer computers –as physical or more often **as virtual machines** -, raw (block) storage, firewalls, load balancers, and networks.

Platform as a service (Paas),

- In the paas model,cloud providers deliver a computing platform and/or solution stack typically including operating system, programming language execution environment, database,and webserver.
- Application developers can develop and run their software solutions on a cloud platform without the cost and complexity of buying and managing the underlying hardware and software layers.

Software as a service(Saas)

- In this model, cloud providers install and operate application software in the cloud and cloud users access the software from cloud client.
- The cloud users do not manage the cloud infrastructure and platform on which the application is running.
- No need to install and run the application on the cloud user's own computers which simplify maintenance and support.

Cloud Deployment MODELs

Three types of Cloud deployment MODELs

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud

Public Cloud

- A form of cloud storage where the enterprise and storage service provider are separate and the data is stored outside of the enterprise's data center.
- With public cloud storage, or external storage clouds, enterprises and small businesses offload their data storage and archival / backup needs to a third-party cloud storage service provider, freeing them from the expensive costs of having to purchase, manage and maintain on-premises storage hardware and software resources.

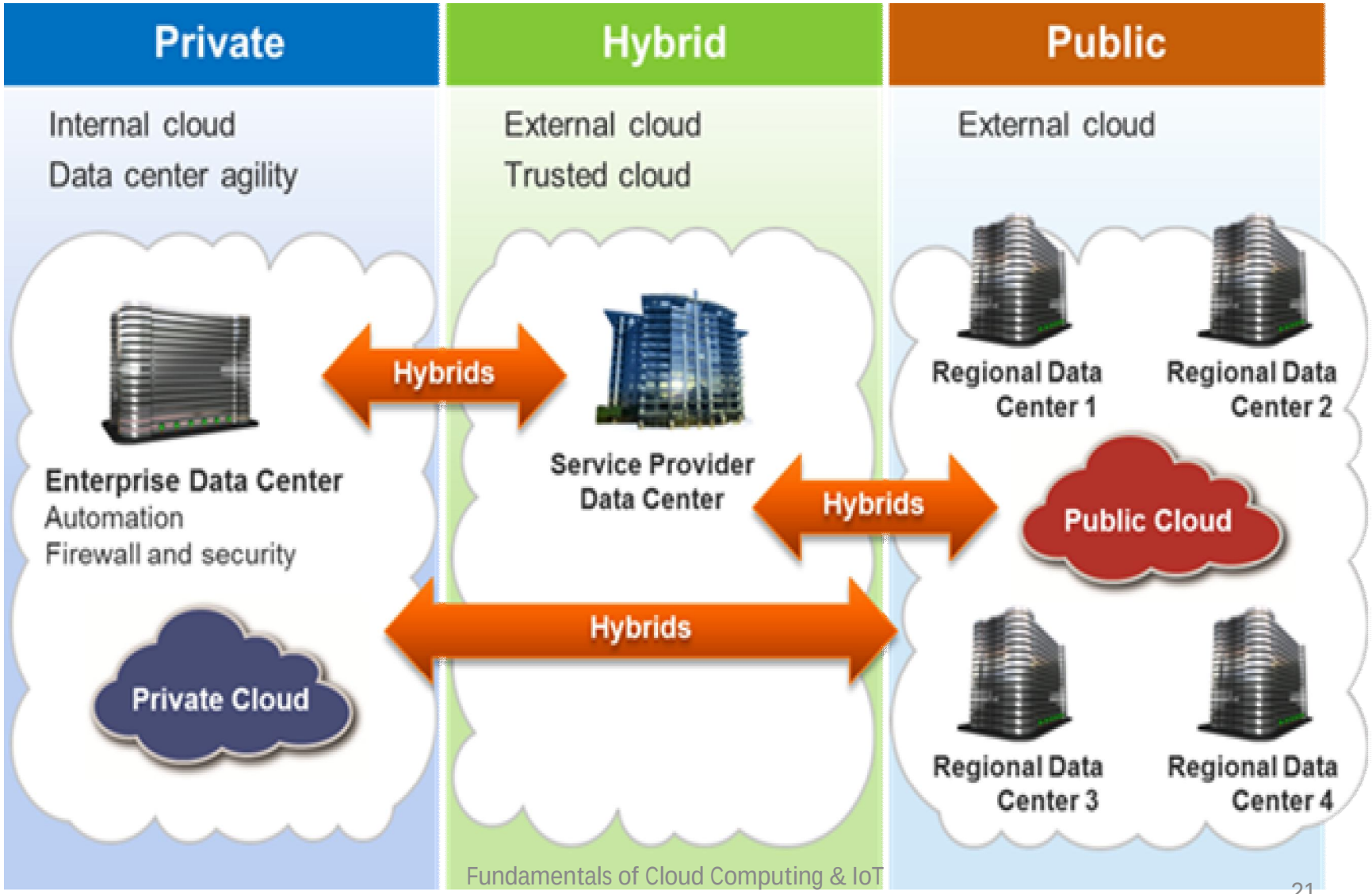
Private Cloud

- The phrase used to describe a cloud computing platform that is implemented within the corporate firewall, under the control of the IT department.
- A private cloud is designed to offer the same features and benefits of public cloud systems, but removes a number of objections to the cloud computing model including control over enterprise and customer data, worries about security, and issues connected to regulatory compliance

Hybrid Cloud

- A combination of public cloud storage and private cloud storage where some critical data resides in the enterprise's private cloud while other data is stored and accessible from a public cloud storage provider.
- Hybrid cloud storage combines the advantages of scalability, reliability, rapid deployment and potential cost savings of public cloud storage with the security and full control of private cloud storage.

Deployment Models



Cloud security & privacy issues

- physical control of the private cloud equipment is more secure than having the equipment off site and under someone else's control .
- Security issues….

1. Sensetive data access
2. Data segregation,
3. Privacy
4. Bug exploitation,
5. Recovery
6. Accountability
7. Malicious insiders
8. Management console security
9. Account control, and
10. multi-tenancy issues

Opportunities issues & challenges

- Security mission critical data
- Privacy of citizen
- Existing infrastructure
- Trust
- Network backbone
- Bandwidth
- Skill of employers

Services and Providers

GOOGLE SITE-license creative commons.

USE XML,HTML,JAVA SCRIPT

*Google doc-SAAS

*WIX-PAAS LICENSE,FREMIUM

*DROPBOX-IAAS

*google APPS

*AMAZON AWS

*IBM BlueMix

PROS and CONS of Cloud Computing

Advantages of Cloud Computing

- Lower Computing Cost
- Improved Performance
- Reduced Software Cost
- Instant Software Updates
- Unlimited Storage Capacity
- Increased Data Reliability
- Device Independence and the “always on!, anywhere and any place”
- Free From Maintenance and the “no-need-to-know”

Cont. …

- Storage and scalability
  - Unlimited storage capability
  - Easily add server and user
- Mobility
  - enables access anywhere with a Web connection
- Cost Efficiency and saving
  - Software cost.
  - Hardware cost.
  - Maintenance
  - Personnel

Cont. …

- Device Diversity and Location Independence
- Backup and Recovery
- Improve performance.
- Latest version availability.
- Easy to deploy
  - Ease to setup
  - Ease to maintain
  - Low speed implementation time.

Disadvantages of Cloud Computing

- Network Connections Dependency
- Data Ownership.
- No Hard Drive.
- Control and Reliability.
- Security risk
- Unpredicted Costs.
- low speed connection?
- Downtime

End of Chapter - 1

Qn. if Any???



Qn If Any?

Cont...
Part I: Cloud Fundamentals
Part II: Cloud Security